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A café network that is secure, reliable and ready to grow

A complete network design for a mid-sized café (30–100 devices): guest and staff Wi-Fi, a point-of-sale (POS) system, CCTV cameras and an internal server. Built on Cisco equipment and validated in simulation.

- 8 isolated VLANs
- Dual internet with automatic failover
- Secure management over SSH
- Ready to expand to new branches

What the network delivers

Four practical outcomes the owner feels every day.

Internet that stays up

Two links from two providers. If the primary drops, the network switches to the backup automatically and returns when the primary recovers.

Protected checkout

POS devices sit in an isolated network and talk only to the POS server and payment gateway.

Isolated guest Wi-Fi

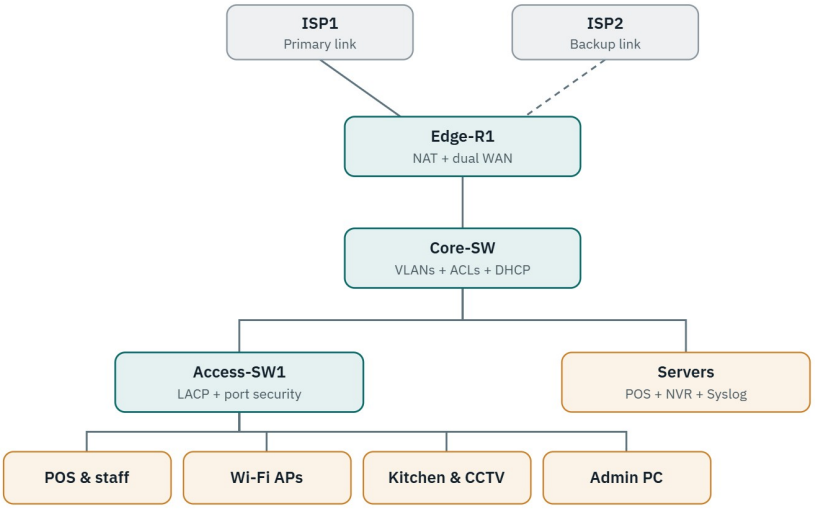
Guests reach the internet only. They cannot reach any device inside the café.

Protected cameras

Cameras talk only to the recorder (NVR) and have no internet access.

Network diagram

From the internet to the devices: each layer has a clear role.



Internal networks (VLANs)

Each group of devices has its own network, with permissions enforced by access control lists (ACLs).

VLAN	Name	Subnet	Access

10	Management	`192.168.10.0/24`	Network devices; access via SSH only
20	Staff	`192.168.20.0/24`	Internet, servers and printers
30	POS	`192.168.30.0/24`	POS server and payment gateway only
40	Servers	`192.168.40.0/24`	POS server and recorder (NVR)
50	Kitchen & printers	`192.168.50.0/24`	Servers only
60	CCTV	`192.168.60.0/24`	Recorder only, no internet
70	Guest Wi-Fi	`192.168.70.0/24`	Internet only
80	Smart devices (IoT)	`192.168.80.0/24`	Internet only, no internal access

Dual internet (failover)

Tested by cutting the primary link: connectivity continued over the backup.

1. **Normal operation**
   All traffic goes through ISP1.
2. **ISP1 goes down**
   The router detects the link failure immediately.
3. **Automatic switchover**
   The backup route through ISP2 takes over with no manual action.
4. **Automatic return**
   When ISP1 recovers, traffic returns to the primary path.

Test results

Every security rule was verified in practice.

Test	Expected	Result
Guest Wi-Fi → internet	Allowed	**Passed**
Guest Wi-Fi → POS server	Blocked	**Blocked**
POS → POS server	Allowed	**Passed**
POS → staff device	Blocked	**Blocked**
Camera → recorder	Allowed	**Passed**
Camera → internet	Blocked	**Blocked**
Staff → printer	Allowed	**Passed**
Primary link cut	Internet continues	**Passed**
SSH from management network	Allowed	**Passed**
SSH from staff network	Blocked	**Blocked**

Device hardening

Layers of protection on management, edge and Layer 2.

Management

- SSH only, Telnet disabled
- Encrypted passwords and a warning banner
- Access from the management network only

Edge

- Spoofed addresses blocked on internet links
- CDP disabled toward the provider

Internal network

- Core acts as the STP root
- DHCP snooping and port security
- Unused ports shut down

Monitoring

- Central syslog on the server
- A log entry for every device change

Expansion plan

The design allows growth without rebuilding.

Inside the café

- Extra switch over LACP
- Additional Wi-Fi access points
- PoE headroom for cameras and APs

New branch

- Same VLAN design
- Branches linked via IPsec / SD-WAN

In production

- FortiGate in place of the router (stateful + IPS)
- High availability for devices
- WLC and RADIUS for central authentication

Simulation limits

This project was built in Cisco Packet Tracer, so access rules (ACLs) are stateless instead of a full firewall, NAT for the second link is a simplified simulation, and Wi-Fi uses two separate access points instead of a WLC. These points improve with real hardware.